

Spectral Nagata Compactification over a Noetherian Base

Abstract

Let Y be a quasi-compact separated locally Noetherian spectral scheme, and let $f: X \rightarrow Y$ be quasi-compact, separated, and locally almost of finite presentation. Then the ∞ -category of compactifications $X \hookrightarrow Z \rightarrow Y$ is cofiltered. This is a rewriting of [1, Chapter 5, Proposition 2.1.6] in the language of spectral algebraic geometry.

1 Statement

Notation 1.1. All rings are connective E_∞ -rings and all schemes are spectral schemes. Write t_0T for the classical truncation of a spectral scheme T , and $T_{\leq n}$ for the scheme obtained by taking the n th Postnikov truncation of its structure sheaf. The terms *separated* and *proper* are those of [4, Section 3.2 and Definition 5.1.2.1]. For spectral schemes, separatedness and properness are detected on classical truncations; see [4, Section 3.2 and Remark 5.1.2.2].

Degree bounds refer to homotopy degrees; coherent modules are bounded and almost perfect. The symbol Map denotes a mapping space. A category is *finitely presented* if it admits a presentation with finitely many objects, generating arrows, and higher coherence cells in total [3, Section 1.2.14].

We use the following form of the spectral Hilbert basis theorem.

Proposition 1.2 (Noetherian finiteness criterion). *Let $h: T \rightarrow V$ be a morphism of spectral schemes, with V locally Noetherian. Then h is locally almost of finite presentation if and only if T is locally Noetherian and t_0h is locally of finite type.*

Proof. This is [4, Proposition 4.2.4.1]; its affine algebraic input is the spectral Hilbert basis theorem [2, Proposition 7.2.4.31]. $\square$

Lemma 1.3 (Finite limits of proper objects). *Let Y be quasi-compact, separated, and locally Noetherian. The full subcategory of Sch_Y spanned by proper morphisms which are locally almost of finite presentation is closed under finite limits.*

Proof. The subcategory contains the terminal object. Every morphism between its objects is proper and locally almost of finite presentation [4, Remark 17.4.1.5], so it is closed under pullbacks. $\square$

Definition 1.4. Let Y be quasi-compact, separated, and locally Noetherian, and let $f: X \rightarrow Y$ be quasi-compact, separated, and locally almost of finite presentation. Regard f as an object of Sch_Y . Define $\mathrm{Comp}_Y(f)$ to be the full subcategory of $(\mathrm{Sch}_Y)_f$ spanned by factorizations

$$X \xrightarrow{j} Z \xrightarrow{p} Y$$

for which j is a quasi-compact open immersion and p is proper and locally almost of finite presentation.

Theorem 1.5 (Spectral compactification). *Under the hypotheses of Definition 1.4, the ∞ -category $\mathrm{Comp}_Y(f)$ is cofiltered. It contains an object for which j is affine. Consequently,*

$$\mathrm{Comp}_Y(f)^{\mathrm{gpd}} \simeq *.$$

The remainder of the paper is devoted to the proof of Theorem 1.5.

Corollary 1.6 (Fixed Noetherian spectral base). *Let R be a connective Noetherian E_∞ -ring, set $S = \mathrm{Spec} R$, and let $\mathcal{C}_S \subseteq \mathrm{Sch}/_S$ be the full subcategory of quasi-compact separated spectral S -schemes locally almost of finite presentation over S . For every morphism $f: X \rightarrow Y$ in $\mathcal{C}_S$, the ∞ -category of factorizations*

$$X \xrightarrow{j} Z \xrightarrow{p} Y, \quad Z \in \mathcal{C}_S,$$

with j open and p proper is cofiltered, and its groupoid completion is contractible.

Proof. The category of factorizations in the statement is canonically isomorphic to $\mathrm{Comp}_Y(f)$, so Theorem 1.5 applies. $\square$

2 Coherent extension of Postnikov classes

Lemma 2.1 (Coherent subsheaf extension). *Let Z be quasi-compact, separated, and locally Noetherian, and let $u: X \hookrightarrow Z$ be a quasi-compact open immersion. For $F \in \mathrm{QCoh}(Z)_{[0,0]}$, every coherent subsheaf $G \subseteq u^*F$ extends to a coherent subsheaf $N \subseteq F$ with $u^*N = G$. Moreover, F is the filtered colimit of its coherent subsheaves. All these subsheaves are concentrated in degree zero.*

Proof. Under $\mathrm{QCoh}(Z)^\heartsuit \simeq \mathrm{QCoh}(t_0 Z)^\heartsuit$, these are [5, Lemmas 28.23.2 and 28.23.3, Tag 01PF and Tag 01PG]; finite-type quasi-coherent sheaves are coherent in the locally Noetherian case [5, Tag 01XZ]. $\square$

Proposition 2.2 (Extension of square-zero thickenings). *Fix a locally Noetherian spectral scheme Y , and work over Y . Let Z be quasi-compact, separated, and locally almost of finite presentation over Y , and let $u: X \hookrightarrow Z$ be affine open. A square-zero extension $i_X: X \hookrightarrow X'$ by a coherent module M_X concentrated in degree $d \geq 0$ extends to a square-zero extension $i_Z: Z \hookrightarrow Z'$ by a coherent module M_Z concentrated in degree d , with induced $u': X' \hookrightarrow Z'$, which is affine open.*

Proof. Proposition 1.2 gives that Z is locally Noetherian. The complex $L_{Z/Y}^{E_\infty}$ is connective and almost perfect because $Z \rightarrow Y$ is locally almost of finite presentation [2, Proposition 7.4.3.9 and Theorem 7.4.3.18]. Since u is affine, Lemma 2.1 expresses $u_*(M_X[-d])$ as a filtered colimit $\mathrm{colim}_\lambda N_\lambda$ with coherent degree-zero N_λ and $u^*N_\lambda \simeq M_X[-d]$. By almost compactness, the adjoint of the classifying map

$$L_{X/Y}^{E_\infty} \longrightarrow M_X[1]$$

factors through $N_\lambda[d+1]$ for some λ [4, Proposition 9.1.5.1], producing a lift

$$L_{Z/Y}^{E_\infty} \longrightarrow M_Z[1]$$

for the coherent module $M_Z = N_\lambda[d] \in \mathrm{QCoh}(Z)_{[d,d]}$ with $u^*M_Z \simeq M_X$. By [4, Section 17.1], this lift gives the required thickening $i_Z: Z \hookrightarrow Z'$. The induced u' is clearly an affine open immersion. $\square$

3 Proof of the compactification theorem

Proof of Theorem 1.5. Write $X_n = X_{\leq n}$.

Classical stage. Nagata compactification [5, Tag 0F41], followed by blowing up a coherent ideal defining the boundary, gives

$$X_0 \xrightarrow{j_0} Z_0 \xrightarrow{p_0} Y \quad (1)$$

with Z_0 classical, j_0 affine open, and p_0 proper [5, Tag 01PH, Tag 02OS, and Tag 07ZU].

Postnikov induction. Suppose $X_n \xrightarrow{j_n} Z_n \xrightarrow{p_n} Y$ has been constructed with Z_n locally Noetherian and n -truncated, j_n affine open, and p_n proper and locally almost of finite presentation. Since $\pi_{n+1}\mathcal{O}_X$ is coherent, [4, Theorem 17.1.3.3] in the relative setting and Proposition 2.2, with $d = n + 1$, give a Cartesian square over Y

$$\begin{array}{ccc} X_n & \hookrightarrow & Z_n \\ \downarrow & & \downarrow \\ X_{n+1} & \hookrightarrow & Z_{n+1}, \end{array}$$

with affine open horizontal maps and $Z_n \hookrightarrow Z_{n+1}$ a square-zero extension by a coherent module concentrated in degree $n + 1$. The fiber sequence of structure sheaves shows that Z_{n+1} is locally Noetherian and $(n + 1)$ -truncated, with $(Z_{n+1})_{\leq n} \simeq Z_n$ over Y . The map p_{n+1} is proper and, by Proposition 1.2, locally almost of finite presentation. Thus $X_{n+1} \hookrightarrow Z_{n+1} \rightarrow Y$ extends the preceding factorization. This completes the induction.

Assembly. By Postnikov convergence [2, Proposition 7.1.3.19], the colimit $p: Z = \operatorname{colim}_n Z_n \rightarrow Y$ exists in Sch/Y , with $Z_{\leq n} \simeq Z_n$ over Y and an induced affine open immersion $j: X \hookrightarrow Z$. For $n \geq i$, $\pi_i\mathcal{O}_Z \simeq \pi_i\mathcal{O}_{Z_n}$ is coherent, so Z is locally Noetherian. Since $t_0p \simeq t_0p_0$, p is proper and locally almost of finite presentation (Proposition 1.2). Thus $(j, p) \in \operatorname{Comp}_Y(f)$.

Cofilteredness. For a diagram $D: K \rightarrow \operatorname{Comp}_Y(f)$ with K a finitely presented category, take its limit $u: X \rightarrow W$ in $(\operatorname{Sch}/Y)_{f/}$. By Lemma 1.3, $W \rightarrow Y$ is proper and locally almost of finite presentation. Proposition 1.2 and [4, Remark 17.4.1.5] show that the existence construction applies to u . It gives $X \hookrightarrow V \rightarrow W$; composition with the limiting cone gives a left cone over D in $\operatorname{Comp}_Y(f)$. This includes $K = \emptyset$, when $W = Y$. Therefore $\operatorname{Comp}_Y(f)$ is cofiltered, and its groupoid completion is contractible by [3, Lemma 5.3.1.20] applied to its opposite. $\square$

References

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